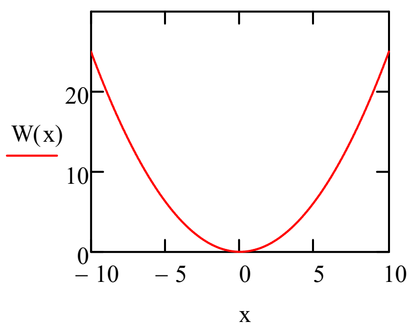
Oscilatorul armonic. Metoda Numerov

$ORIGIN = 0$

$h := 6.625\cdot 10^{-34}$ $m := 0.067\cdot 9.1\cdot 10^{-31}$ $ee := 1.6\cdot 10^{-19}$

$$hp := \frac{h}{2\cdot\pi}$$

$$W(x) := \frac{x^2}{4}$$



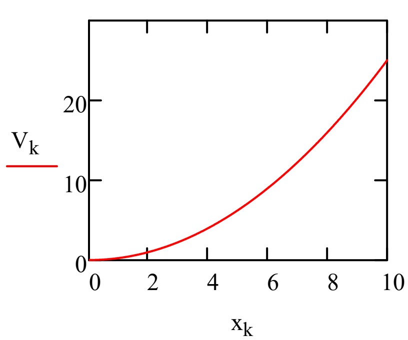
Discretizarea coordonatei

$N := 1000$ $dx := 0.01$

$k := 0 .. N$

$x_k := dx\cdot k$

$V_k := W(x_k)$



Discretizarea energiei

$M := 5000$

$i := 0 .. M$

$E_i := i \cdot 0.003 \qquad \max(E) = 15$

Metoda Numerov

$g_{i,k} := \left(E_i - V_k\right)$

$f_{i,k} := 1 + g_{i,k} \cdot \frac{dx^2}{12}$

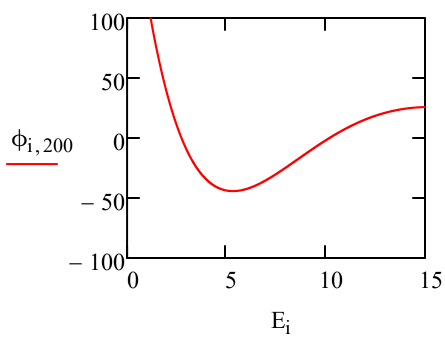
$j := 1 .. N - 1$

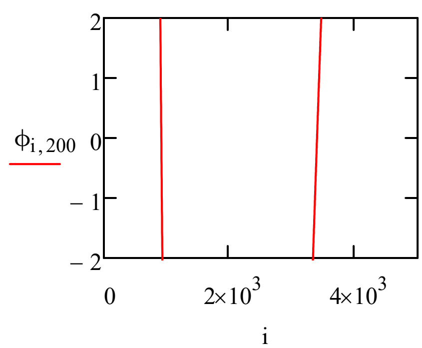
Solutii pare

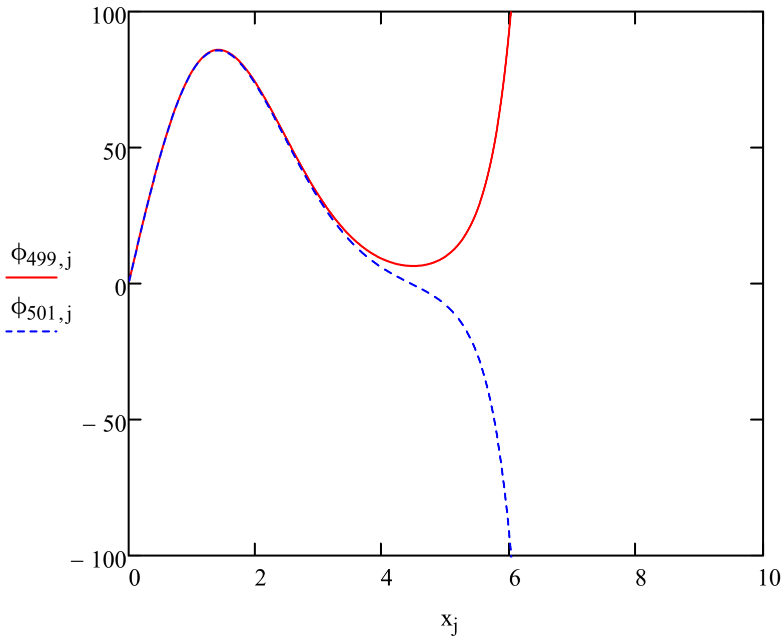
$\phi_{i,0} := 0$

$\phi_{i,1} := 1$

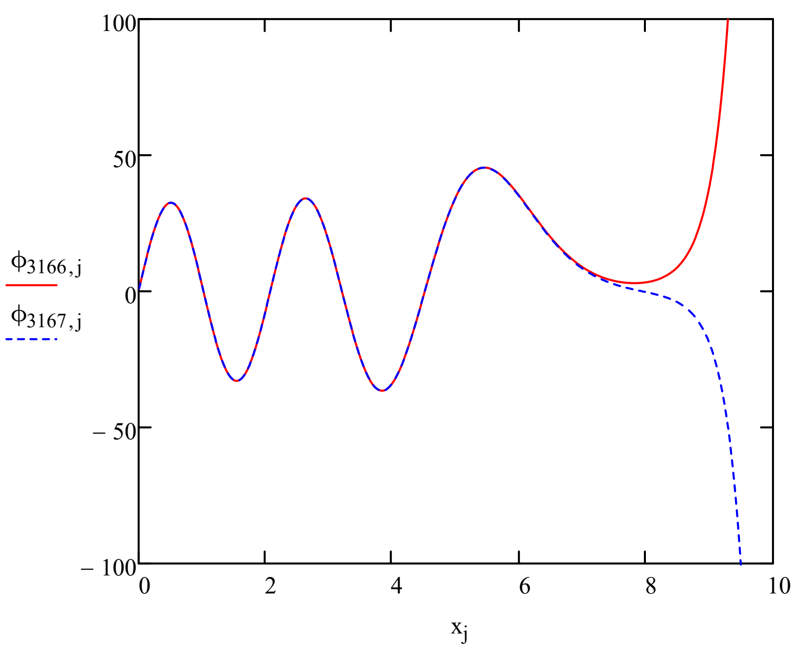
$$\phi_{i,j+1} := \frac{\left(12 - 10 \cdot f_{i,j}\right) \cdot \phi_{i,j} - f_{i,j-1} \cdot \phi_{i,j-1}}{f_{i,j+1}}$$







$E_{500} = 1.5$



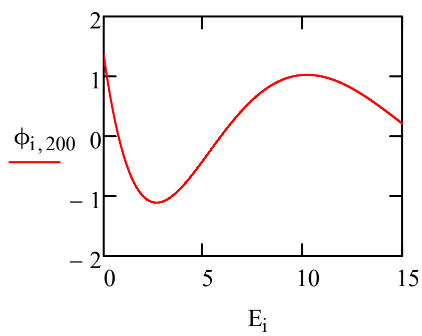
$E_{3167} = 9.501$

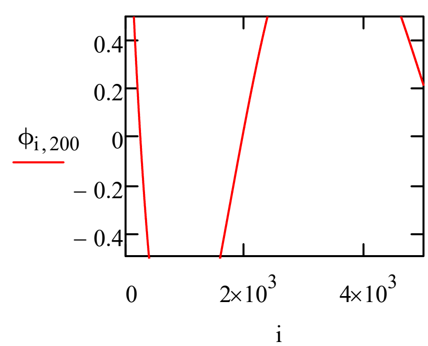
Solutii impare

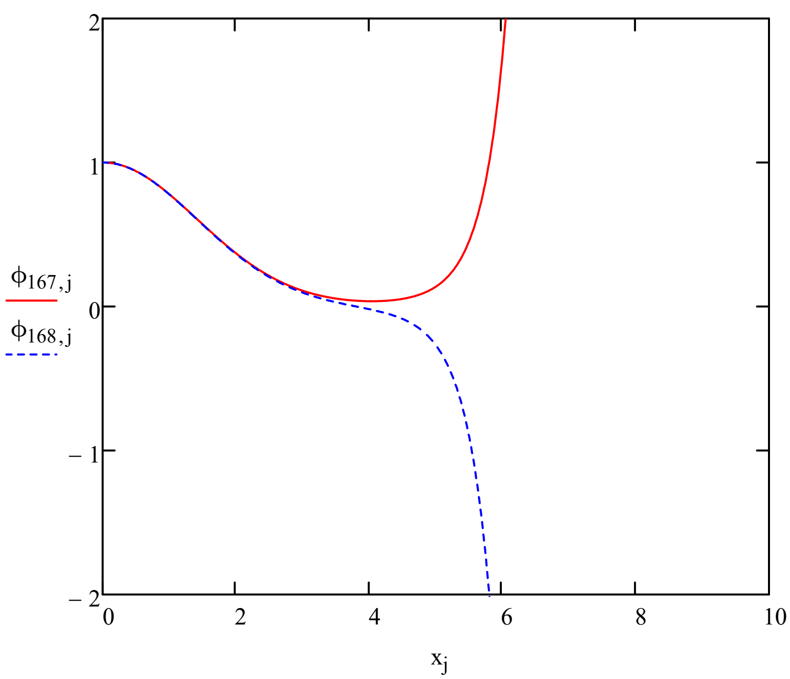
$$\phi_{i,0} := 1$$

$$\phi_{i,1} := 1$$

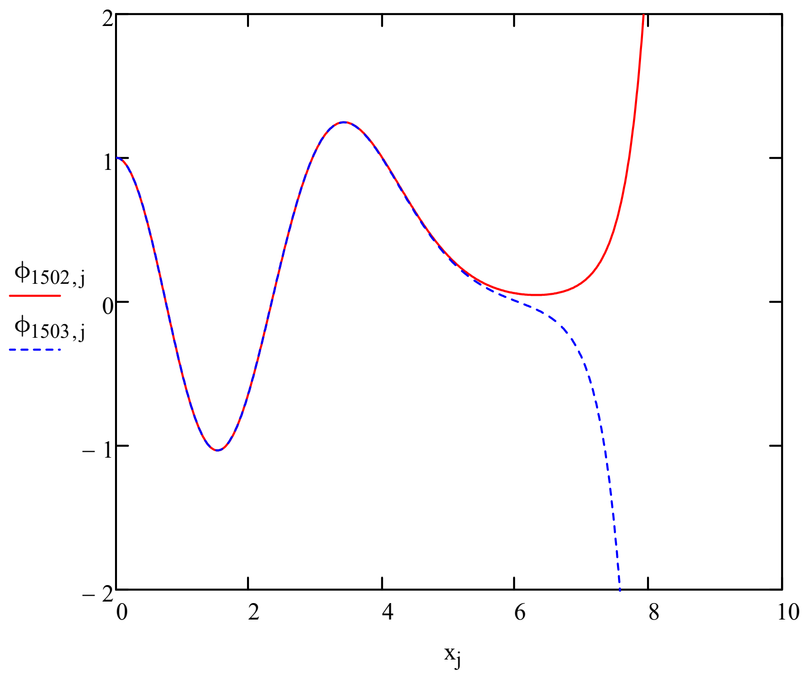
$$\phi_{i,j+1} := \frac{\left(12 - 10 \cdot f_{i,j}\right) \cdot \phi_{i,j} - f_{i,j-1} \cdot \phi_{i,j-1}}{f_{i,j+1}}$$



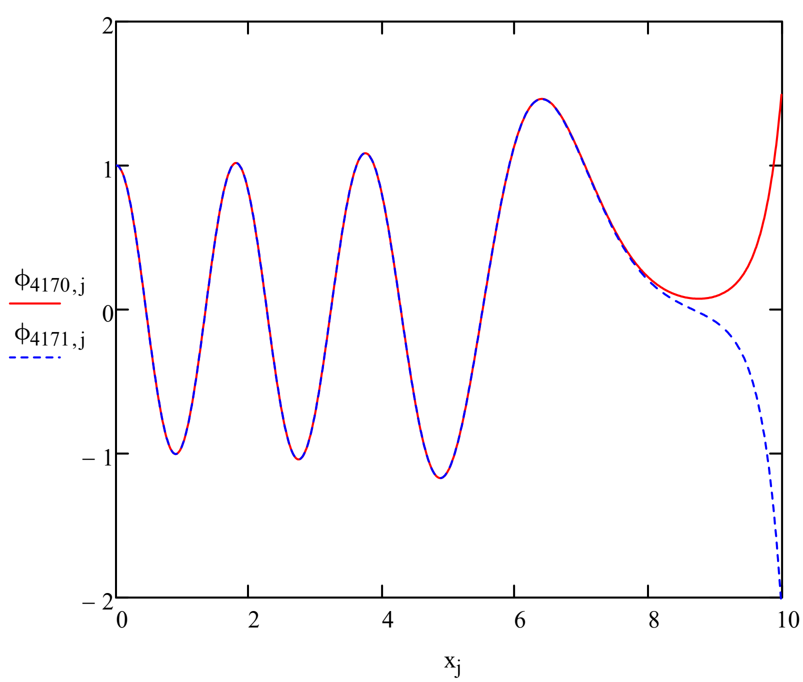




$E_{167} = 0.501$



$E_{1833} = 5.499$



$E_{4167} = 12.501$